Dipole Elements, huh?

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1 What's the idea here

From the Oregon manuscript, we have the spontaneous Raman scattering rate (eq 41 in the version I have)

$$\Gamma_{i\to j,spon} = \frac{E_0 e^4 (\omega_{laser} - \omega_f)^3}{12\pi c^3 \epsilon_0 \hbar^3} \sum_q \left| \sum_k \left(\frac{\langle f|\boldsymbol{D}_q|k\rangle\langle k|\boldsymbol{D}^*_{laser}|i\rangle}{\omega_k - \omega_{laser}} + \frac{\langle f|\boldsymbol{D}^*_{laser}|k\rangle\langle k|\boldsymbol{D}_q|i\rangle}{\omega_k + \omega_{laser} - \omega_f} \right) \right|^2 \tag{1}$$

here, $\boldsymbol{D}_x = \vec{d} \cdot \hat{e}_x$ represents the dipole operator ($\boldsymbol{D}^*_x = e\vec{r} \cdot \hat{e}^*_x$).

A typical calculation of transition strengths involves the relation between the spontaneous decay rate from excited state $|e\rangle$ to lower state $|g\rangle$, Γ_{ge},

$$\Gamma_{ge} = \frac{\omega^3}{3\pi\epsilon_0\hbar c^3} \frac{2J_g + 1}{2J_e + 1} |\langle J_g||\boldsymbol{D}||J_e\rangle|^2 \tag{2}$$

It is important to observe a few points.

1. There are multiple conventions for defining the reduced dipole element. The decay rate is less ambiguous. The two common conventions differ by a $\sqrt{2J+1}$ factor

   - Edmonds, Zare, Brown and Carrington

   $$\langle Jm_J|\boldsymbol{D}_q|J'm'_J\rangle = \frac{1}{\sqrt{2J+1}} {}_{BC}\langle J||\boldsymbol{D}||J'\rangle_{BC} \langle J'm'_J; 1q|Jm_J\rangle \tag{3}$$

   - Brink and Satchler, Steck, Me for today

   $$\langle Jm_J|\boldsymbol{D}_q|J'm'_J\rangle = \langle J||\boldsymbol{D}||J'\rangle \langle Jm_J|J'm'_J; 1q\rangle \tag{4}$$

2. The reduced matrix element is not equal to it's own conjugate–the ordering of the primed (J') and unprimed (J) are important.

   - Following Steck:

   $$\langle J'||\boldsymbol{D}||J\rangle = (-1)^{J'-J} \sqrt{\frac{2J+1}{2J'+1}} \langle J||\boldsymbol{D}||J'\rangle^* \tag{5}$$

- Following Brown and Carrington

$$ {}_{BC}\langle J'||\boldsymbol{D}||J\rangle_{BC} = (-1)^{J'-J}\, {}_{BC}\langle J||\boldsymbol{D}||J'\rangle^*_{BC} \tag{6}$$

3. A calculation of the reduced matrix element following Eq (2) drops a phase factor, really just a $(-1)^j, j \in \mathbb{Z}$ factor. This factor is calculated by reducing to the $\langle L||\boldsymbol{D}||L'\rangle$ matrix elements.

2 deets

Let's look at the following matrix element, expressed with a reduced matrix element through the Weigner-Eckart theorem:

$$\langle Fm_F|D_q|F'm'_F\rangle = \langle F||\boldsymbol{D}||F'\rangle\langle Fm_F|F'm'_F;1q\rangle \tag{7}$$

Switching primed and unprimed terms and flipping sign of q gives

$$\langle F'm'_F|D_{-q}|Fm_F\rangle = \langle F'||\boldsymbol{D}||F\rangle\langle F'm'_F|Fm_F;1-q\rangle \tag{8}$$

To switch the ordering of the Clebsch-Gordon symbol, I use the following

$$\langle Fm_F|F'm'_F1-q\rangle = (-1)^{F'-F+m'_F-m_F}\sqrt{\frac{2F+1}{2F'+1}}\langle F'm'_F|Fm_F1q\rangle$$

or correspondingly,

$$\langle F'm'_F|Fm_F1q\rangle = (-1)^{F'-F+m'_F-m_F}\sqrt{\frac{2F'+1}{2F+1}}\langle Fm_F|F'm'_F1-q\rangle$$

Now going back to the Weigner-Eckart expression

$$\begin{aligned}\langle F'm'_F|D_q|Fm_F\rangle &= \langle F'||\boldsymbol{D}||F\rangle\langle F'm'_F|Fm_F;1q\rangle \\ &= \langle F'||\boldsymbol{D}||F\rangle(-1)^{F'-F+m'_F-m_F}\sqrt{\frac{2F'+1}{2F+1}}\langle Fm_F|F'm'_F1-q\rangle\end{aligned} \tag{9}$$

Now, via the spectator theorem stating that for F = I + J, I is a spectator, we reduce the F matrix element

$$\langle F'||\boldsymbol{D}||F\rangle = \langle J'||\boldsymbol{D}||J\rangle(-1)^{F+J'+1+I}\sqrt{(2F+1)(2J'+1)}\begin{Bmatrix} J' & J & 1 \\ F & F' & I\end{Bmatrix} \tag{10}$$

Plugging eq (10) into eq (9),

$$\langle F'm'_F|D_q|Fm_F\rangle = \langle J'||\boldsymbol{D}||J\rangle\langle Fm_F|F'm'_F1-q\rangle(-1)^{F+J'+1+I}\sqrt{(2F+1)(2J'+1)}\begin{Bmatrix} J' & J & 1 \\ F & F' & I \end{Bmatrix}$$
$$* (-1)^{F'-F+m'_F-m_F}\sqrt{\frac{2F'+1}{2F+1}}$$
$$= \langle J'||\boldsymbol{D}||J\rangle(-1)^{J'+F'+1+I+m'_F-m_F}\sqrt{(2J'+1)(2F'+1)}\begin{Bmatrix} J' & J & 1 \\ F & F' & I \end{Bmatrix}$$
$$* \langle Fm_F|F'm'_F1-q\rangle \tag{11}$$

In order to reverse the J dipole element, I use eq (5). Inserting this, I obtain the following result:

$$\langle F'm'_F|D_q|Fm_F\rangle = \langle J||\boldsymbol{D}||J'\rangle^* - 1)^{J'-J}(-1)^{J'+F'+1+I+m'_F-m_F}$$

$$* \sqrt{(2J'+1)(2F'+1)}\begin{Bmatrix} J' & J & 1 \\ F & F' & I \end{Bmatrix}\langle Fm_F|F'm'_F1-q\rangle \tag{12}$$

$$= \langle J||\boldsymbol{D}||J'\rangle^*(-1)^{2J'-J+F'+1+I+m'_F-m_F}$$

$$* \sqrt{(2J'+1)(2F'+1)}\begin{Bmatrix} J' & J & 1 \\ F & F' & I \end{Bmatrix}\langle Fm_F|F'm'_F1-q\rangle \tag{13}$$

Now, I return to the conjugate of this matrix element $\langle F'm'_F|D_q|Fm_F\rangle$. first applying Weigner-Eckart, eg (7):

$$\langle Fm_F|D_q|F'm'_F\rangle = \langle F||\boldsymbol{D}||F'\rangle\langle Fm_F|F'm'_F;1q\rangle \tag{14}$$

And additionally applying the I-spectator theorem, eq (10):

$$\langle F||\boldsymbol{D}||F'\rangle = \langle J||\boldsymbol{D}||J'\rangle(-1)^{F'+J+1+I}\sqrt{(2F'+1)(2J+1)}\begin{Bmatrix} J & J' & 1 \\ F' & F & I \end{Bmatrix} \tag{15}$$

By combination of eq (14) and (15), I arrive at

$$\langle Fm_F|D_q|F'm'_F\rangle = \langle J||\boldsymbol{D}||J'\rangle\langle Fm_F|F'm'_F;1q\rangle(-1)^{F'+J+1+I}\sqrt{(2F'+1)(2J+1)}\begin{Bmatrix} J & J' & 1 \\ F' & F & I \end{Bmatrix} \tag{16}$$

Now I'd like to compare (16) to our conjugate expression eq (13). Note that I am free to permute columns of the 6j-symbol, due to the symmetries of this symbol. Here, I swap column 1 and 2 of the 6j in eq (13):

$$\langle F'm'_F|D_q|Fm_F\rangle = \langle J||\boldsymbol{D}||J'\rangle^*\langle Fm_F|F'm'_F1-q\rangle(-1)^{2J'-J+F'+1+I+m'_F-m_F}$$

$$* \sqrt{(2J'+1)(2F'+1)}\begin{Bmatrix} J & J' & 1 \\ F' & F & I \end{Bmatrix} \tag{17}$$

By comparison of (16) and (17), with the physical assumption that the matrix element $\langle J||\boldsymbol{D}||J'\rangle$ is real-valued, I find the following relationship

$$\langle F'm'_F|D_q|Fm_F\rangle = (-1)^{2J'-2J+m'_F-m_F}\langle Fm_F|D_{-q}|F'm'_F\rangle \tag{18}$$

Notice the change in sign of the photon polarization ($q \to -q$).

3 Using this to calculate Γ

Now, consider eq (2);

$$\Gamma_{\Gamma,\Gamma'} = \frac{\omega^3}{3\pi\epsilon_0\hbar c^3}\frac{2J+1}{2J'+1}\left|\langle J||\boldsymbol{D}||J'\rangle\right|^2$$

From this, we solve for the reduced matrix element in terms of the excited state decay rate from J' to J.

$$|\langle J||\boldsymbol{D}||J'\rangle| = \sqrt{\frac{3\pi\epsilon_0\hbar c^3}{\omega}\frac{2J'+1}{2J+1}\Gamma_{J,J'}} \tag{19}$$

Given the expression (19), I would like to express the full scattering rate equation (1) in terms of dipole elements with this ordering. I'll write the two terms of the inner sum as "term A" and "term B". Note the minus signs on the RHS refers to the change in sign of the photon polarization ($\sigma_\pm \to \sigma_\mp$).

Term A:

$$\frac{\langle f|\boldsymbol{D}_q|k\rangle\langle k|\boldsymbol{D}^*_{laser}|i\rangle}{\omega_k-\omega_{laser}} = (-1)^{2J_k-2J_i+m_{Fk}-m_{Fi}}\frac{\langle f|\boldsymbol{D}_q|k\rangle\langle i|\boldsymbol{D}^*_{-laser}|k\rangle}{\omega_k-\omega_{laser}} \tag{20}$$

Term B:

$$\frac{\langle f|\boldsymbol{D}^*_{laser}|k\rangle\langle k|\boldsymbol{D}_q|i\rangle}{\omega_k+\omega_{laser}-\omega_f} = (-1)^{2J_k-2J_i+m_{Fk}-m_{Fi}}\frac{\langle f|\boldsymbol{D}^*_{laser}|k\rangle\langle i|\boldsymbol{D}_{-q}|k\rangle}{\omega_k+\omega_{laser}-\omega_f} \tag{21}$$

With the E1-transition selection rules for $\Delta J = 0, \pm 1$, the $(-1)^{2J_k-2J_i+m_{Fk}-m_{Fi}}$ factor reduces to $(-1)^{m_{Fk}-m_{Fi}}$. Now, to re-express the full scattering rate with the dipole elements ordered for straightforward implementation in terms of the linewidth equation (19)

$$\Gamma_{i\to f,spon} = \frac{E_0 e^4(\omega_{laser}-\omega_f)^3}{12\pi c^3\epsilon_0\hbar^3}\sum_q$$
$$\left|\sum_k\left((-1)^{2J_k-2J_i+m_{Fk}-m_{Fi}}\frac{\langle f|\boldsymbol{D}_q|k\rangle\langle i|\boldsymbol{D}^*_{-laser}|k\rangle}{\omega_k-\omega_{laser}} + (-1)^{2J_k-2J_i+m_{Fk}-m_{Fi}}\frac{\langle f|\boldsymbol{D}^*_{laser}|k\rangle\langle i|\boldsymbol{D}_{-q}|k\rangle}{\omega_k+\omega_{laser}-\omega_f}\right)\right|^2 \tag{22}$$

Here's the simplified form. This result does not explicitly contain the reduced dipole element, and is independent of the convention used to define $\langle J||\boldsymbol{D}||J'\rangle$. This will hold true for reduced dipole elements defined with the Brown and Carrington convention, as well:

$$\Gamma_{i\rightarrow f,spon} = \frac{E_0 e^4(\omega_{laser}-\omega_f)^3}{12\pi c^3\epsilon_0\hbar^3}$$
$$\sum_q \left|\sum_k \left((-1)^{m_{F\,k}-m_{F\,i}}\frac{\langle f|\boldsymbol{D}_q|k\rangle\langle i|\boldsymbol{D}^*_{-laser}|k\rangle}{\omega_k-\omega_{laser}} + (-1)^{m_{F\,k}-m_{F\,i}}\frac{\langle f|\boldsymbol{D}^*_{laser}|k\rangle\langle i|\boldsymbol{D}_{-q}|k\rangle}{\omega_k+\omega_{laser}-\omega_f}\right)\right|^2 \tag{23}$$

This formula can be computed directly with the evaluation of matrix elements using eq (16) in combination with eq (19), where all dipole moments are ordered so that the reduced matrix element will be correctly determined via (19) simply using A-coefficients for the decay rates. I have not explicitly included the phase factor described in section 1, statement 3.